

BRIDGE2AI

Building Effective Interdisciplinary Teams

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The Foundation: Why Interdisciplinarity matters

Complex scientific problems demand diverse perspectives:

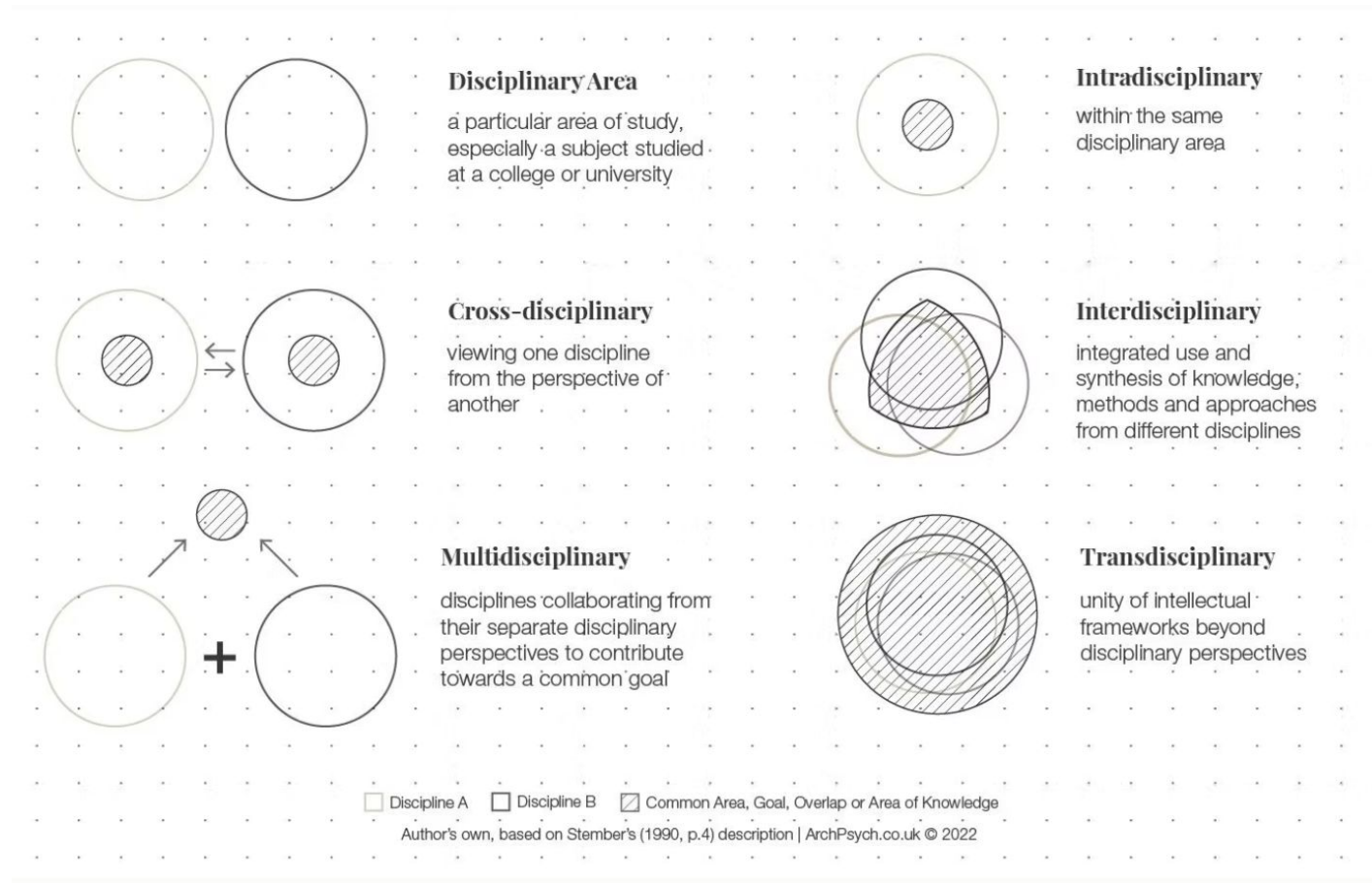
- Diverse teams outperform homogeneous teams on complex problems (Page 2007)
- But multidisciplinary alone isn't enough - inclusion practices determine whether engaging a broader scope of disciplines helps or hurts (Nishii 2013)
- Small changes in process can have big impacts on who participates and how (Woolley 2010)



Levels & intersections of heterogeneity

- Surface-Level Heterogeneity
 - Demographics: gender, race, age, nationality
 - Disciplinary backgrounds
 - Institutional affiliations
 - Career stages
- Deep-Level Heterogeneity
 - Thinking styles (analytical vs. intuitive)
 - Work preferences (individual vs. collaborative)
 - Communication styles (direct vs. indirect)
 - Risk tolerance (conservative vs. experimental)
 - Skillsets (technical vs. interpersonal)

Disciplinary Continuum

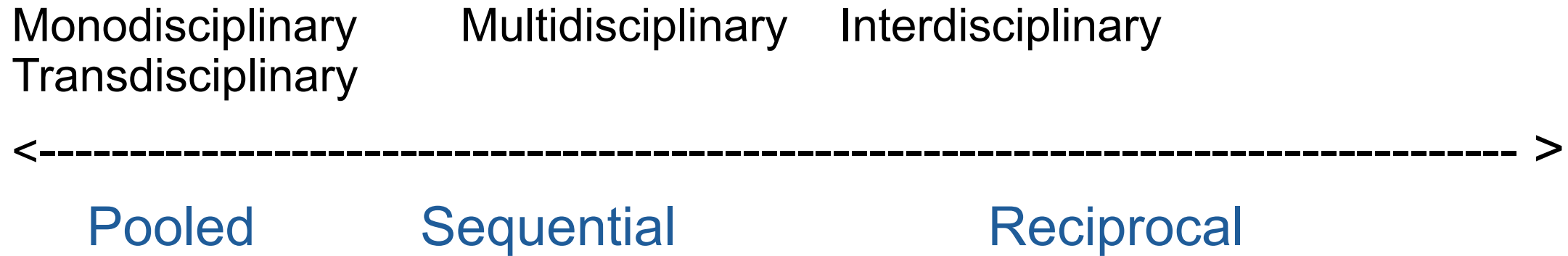


(Manolakelli, A. 2022)

Thompson's Levels of Interdependence & Coordination

Type of Interdependence		Key Coordination Method	
Pooled	Team members work independently and contribute their separate pieces to a common goal. The output is the sum of their efforts.	Standardization	Relies on establishing common rules, procedures, and standards that all members follow. This ensures outputs are consistent and can be easily aggregated with minimal interaction.
Sequential	Work flows in an assembly-line fashion. One person's or group's output becomes the input for the next person or group.	Planning & Scheduling	Focuses on creating schedules and timelines to manage the flow of work between stages. The goal is to ensure smooth and timely hand-offs from one member to the next.
Reciprocal	Work is iterative and requires back-and-forth interaction. Each member's output is an input for others, and vice-versa, in a dynamic process.	Mutual Adjustment	Requires constant communication, feedback, and shared decision-making. Team members continuously adapt their actions in response to others. This is the most intensive and demanding form of coordination.

Overlay of Thompson on Continuum



Individual Reflection (Exercise)

- What kind of team are you?
- Where does your team fall on the continuum? Where would you like it to be?
- How does your work connect to others on the team?
- Does your level of interdependence match team type?
- Do your team's processes align with its level of interdependence?

Frameworks For Success

The structural foundation (Hackman's 5 Conditions)

1. Real Team: Clearly bounded, stable membership.
2. Compelling Direction: A clear, challenging, and consequential purpose.
3. Enabling Structure: The right mix of people and clear norms of conduct.
4. Supportive Context: Access to resources, information, and rewards for teamwork.
5. Expert Coaching: Help with process and removing roadblocks.

Team Dynamics (Amy Edmondson's Principles):

- Psychological Safety: A shared belief that the team is safe for interpersonal risk-taking. People feel comfortable speaking up with ideas, questions, or mistakes.
- Inclusive Leadership: Leaders who actively invite and appreciate contributions from everyone.
- Deliberate Processes: Intentionally designing how the team communicates, makes decisions, and resolves conflict.

Group Exercise

Take a few minutes and rank your team based on foundational principles.

Next, share your rankings with your neighbor, take turns, and discuss the following:

- Why did you choose these rankings?
- What are some things the team is doing well?
- Where is there an area for growth?
- Are there specific challenges that you see?

If you could pick one area to work on, psychological safety, inclusive leadership, or deliberate processes which one would you choose? Why?

References

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